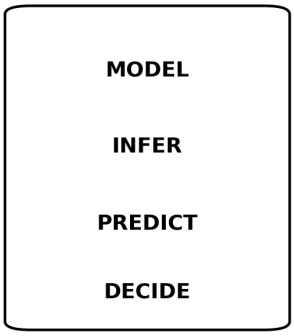
Class 01 — Course Orientation and the Probabilistic View of Advanced Machine Learning

Advanced Machine Learning for Artificial Intelligence

Class 01 · Course Orientation and the Probabilistic View

Prof. Wonsang You · Data Science · Dongduk Women's University · Fall 2026

$$p(\theta, z \mid \mathcal{D}) \propto p(\mathcal{D} \mid z, \theta)\, p(z \mid \theta)\, p(\theta)$$



> **Advanced Machine Learning for Artificial Intelligence**
> **Prof. Wonsang You, Data Science, Dongduk Women's University**
> **Fall Semester, 2026 · DSC1002 · 3 credits**
> **Schedule update, effective September 8:** Tuesday, 13:00–15:00 (one continuous meeting per week)
> **Today's shortened meeting:** approximately 40 minutes

Learning objectives

By the end of this class, you should be able to:

1. Explain the scope, structure, prerequisites, and assessment policy of this course.
2. Distinguish four broad goals of advanced machine learning: **prediction, generation, discovery, and action**.
3. Explain why uncertainty should be represented by probability distributions rather than only by point estimates.
4. Recall the mathematical roles of conditional probability, Bayes' rule, expectation, variance, likelihood, MLE, MAP, gradients, Hessians, and stochastic gradient descent.
5. Describe the recurring workflow that connects **probability, statistics, optimization, prediction, and decision making**.

1. Course at a glance

Course purpose

This course studies advanced machine learning for artificial intelligence from a **probabilistic, model-based perspective**. Rather than treating machine learning only as fitting a function from inputs to outputs, we will study how to:

- represent uncertainty about data, latent variables, parameters, and predictions;
- infer hidden structure from observations;
- construct predictive and generative models;
- reason about distribution shift and out-of-distribution inputs;
- make decisions under uncertainty;
- connect statistical modeling, deep learning, reinforcement learning, and causality.

The course is theory-oriented, but Python-based assignments will require you to **derive, implement, test, and analyze** important algorithms.

Expected background

Students are recommended to have prior exposure to:

- Python programming;
- linear algebra;
- probability and statistics;
- optimization;
- basic machine learning.

A laptop or equivalent personal computing device is required for coding assignments and possible in-class practice.

Textbooks and resources

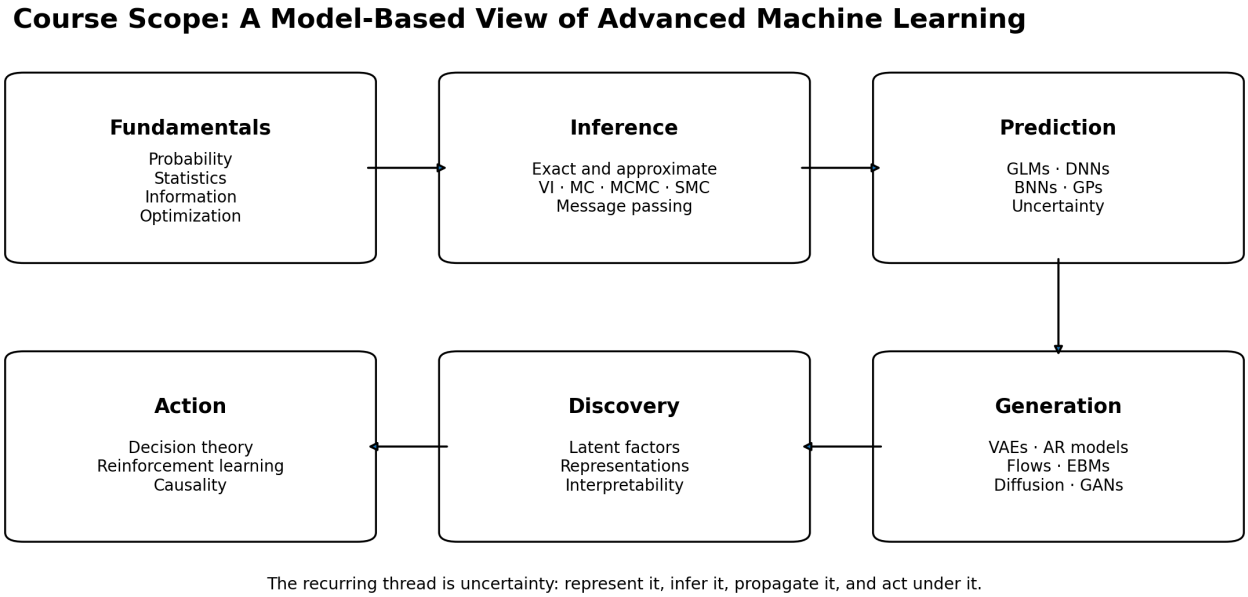
Role	Resource	Use in this course
Main textbook	Kevin P. Murphy, *Probabilistic Machine Learning: Advanced Topics*, MIT Press, 2023	Main source for advanced models and inference methods
Supplementary textbook	Kevin P. Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press, 2022	Review of probability, statistics, linear algebra, optimization, and basic ML
Supplementary material	Kevin P. Murphy, *Supplementary Material for Probabilistic Machine Learning: Advanced Topics*	Optional deeper examples and extensions
Official code	probml/pyprobml — Book 2 notebooks	Reproducible Python/JAX demonstrations
Course repository	lunalab-ai/advML	Course notebooks, assignments, and supporting files

Source alignment for Class 01

Class topic	Textbook alignment	Optional notebook
Probabilistic view of advanced ML	Main textbook, Chapter 1, pp. 1–2	Book 2 notebook directory
Probability review	Supplementary textbook, Chapter 2, especially §§2.1–2.3	Intro to probability — Colab
Statistics, MLE, MAP, regularization	Supplementary textbook, Chapter 4, especially §§4.1–4.5	Introduction to Bayesian statistics — Colab
Optimization, GD, and SGD	Supplementary textbook, Chapter 8, especially §§8.1–8.4	Steepest descent demo — Colab · Optimization with JAX — Colab

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2. How the semester is organized

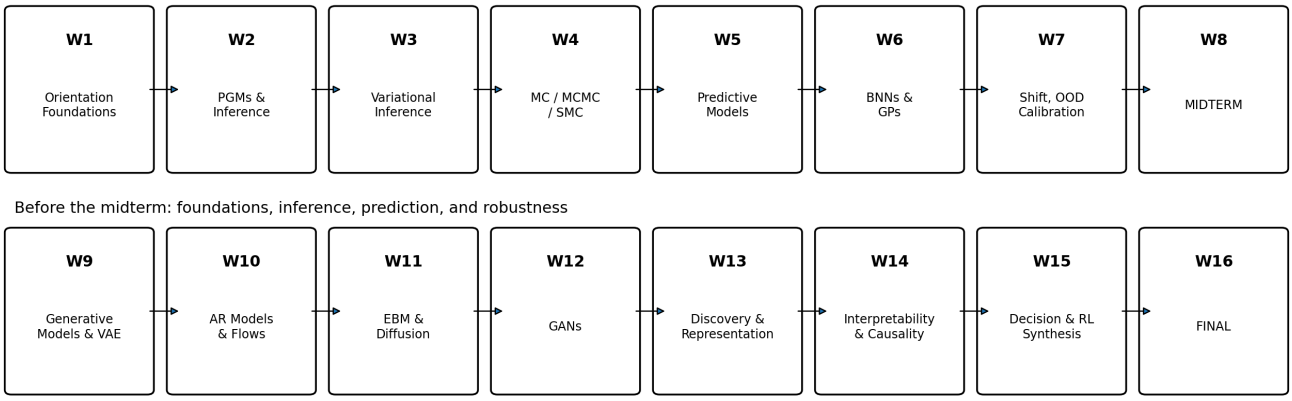


The recurring thread is uncertainty: represent it, infer it, propagate it, and act under it.

The main textbook is organized around six large blocks. The course follows the same conceptual progression, with some compression and reordering.

1. **Fundamentals:** probability, statistics, graphical models, information, and optimization.
2. **Inference:** exact and approximate inference, message passing, variational inference, Monte Carlo, MCMC, and SMC.
3. **Prediction:** generalized linear models, deep networks, Bayesian neural networks, Gaussian processes, calibration, and distribution shift.
4. **Generation:** VAEs, autoregressive models, normalizing flows, energy-based models, diffusion models, and GANs.
5. **Discovery:** latent factor models, representation learning, and interpretability.
6. **Action:** decision making under uncertainty, reinforcement learning, and causality.

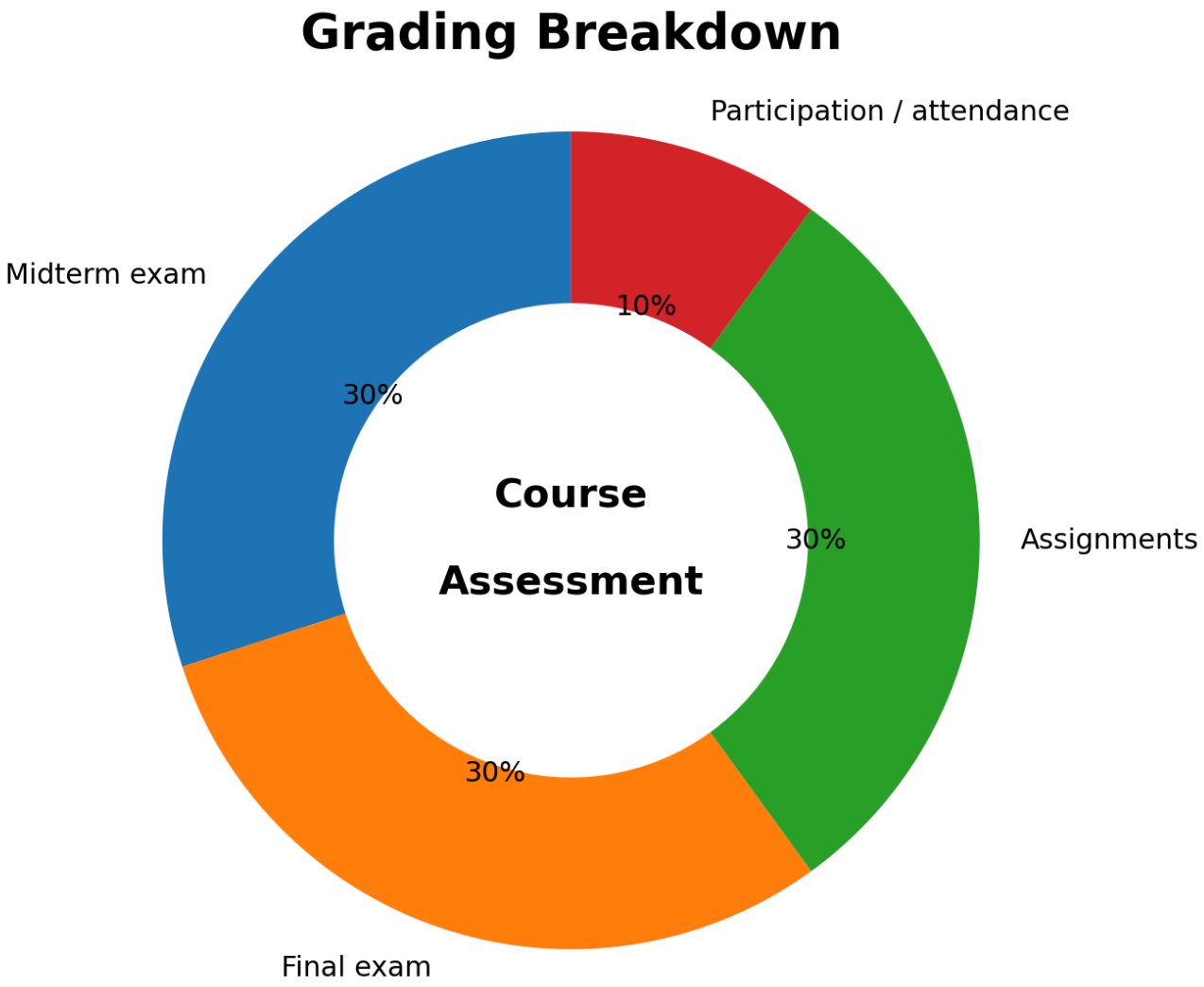
Sixteen-Week Roadmap



Weekly plan

Week	Main topic
1	Course overview; probabilistic perspective; review of probability, statistics, and optimization
2	Probabilistic graphical models; inference overview; conditional independence; message passing
3	Variational inference; ELBO; mean-field approximation; probabilistic optimization
4	Monte Carlo; importance sampling; MCMC; sequential Monte Carlo
5	Predictive models; generalized linear models; deep neural networks as probabilistic predictors
6	Bayesian neural networks; Gaussian processes; predictive uncertainty; function-space views
7	Beyond iid; distribution shift; calibration; out-of-distribution issues
8	**Midterm examination**
9	Generative models; latent variable models; variational autoencoders
10	Autoregressive models; normalizing flows; likelihood-based generation
11	Energy-based models; diffusion; score-based and denoising-based generation
12	GANs; adversarial learning; comparison of major generative model families
13	Latent factor models; discovery; representation learning
14	Interpretability; feature attribution; counterfactual explanations; basic causal inference
15	Decision making under uncertainty; reinforcement learning; course synthesis
16	**Final examination**

3. Grading and academic policies



Component	Weight	What it evaluates
Midterm examination	30%	Mathematical and conceptual understanding of the first half
Final examination	30%	Integrated understanding of the second half and the whole course
Assignments	30%	Derivation, implementation, experimentation, analysis, reading, and technical communication
Participation and attendance	10%	Attendance, preparation, engagement, and contribution to class activities

Assignment policy

Assignments may include:

- mathematical derivations;
- implementation of algorithms;
- experimental design and analysis;

- short research-paper readings;
- concise concept summaries.

Late submission, plagiarism, or unauthorized copying of code may result in a penalty or a grade of zero. Assignment-specific requirements and submission procedures will be announced with each assignment.

Attendance policy

- Two late arrivals count as one absence.
- Under university regulations, a student whose total absences exceed **one-fifth of the total class hours** receives an F grade.

> **Working principle:** Do not aim only to obtain a numerical result. You should be able to explain the model assumptions, derive the objective, justify the algorithm, inspect failure modes, and communicate what the experiment means.

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4. Why a probabilistic perspective?

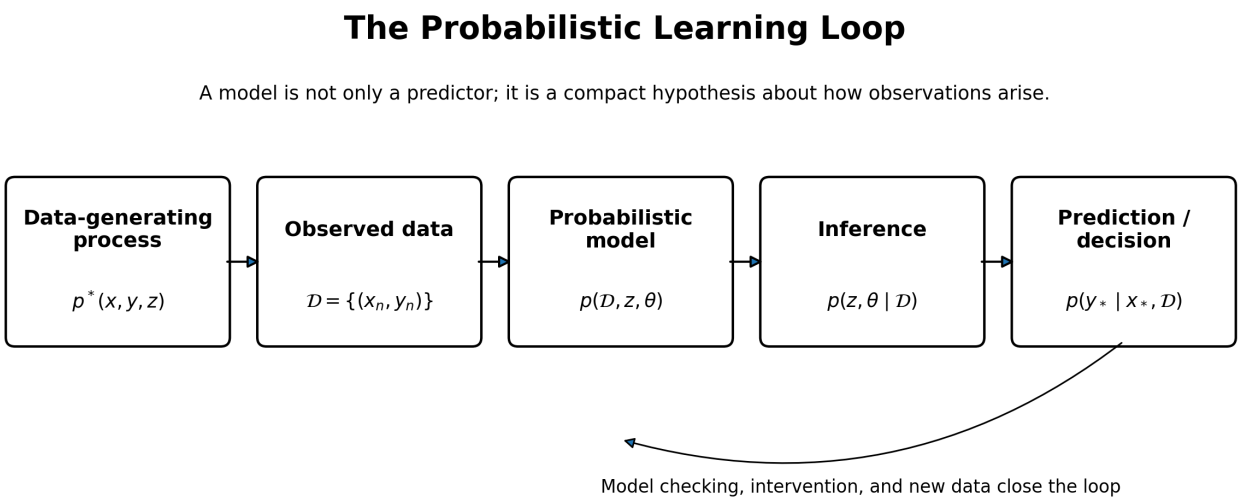
A standard predictive model is often written as a function

$$\hat{y} = f(x; \theta).$$

This representation is useful, but it leaves several important questions unanswered:

- How uncertain is the prediction?
- Is uncertainty caused by irreducible noise or by insufficient knowledge?
- What hidden variables could have generated the observation?
- How should multiple plausible models be combined?
- How should we choose an action when errors have different costs?
- What changes when the test distribution differs from the training distribution?

A probabilistic model answers such questions by describing a **distribution over possible values**, not only one value.



4.1 Unknown quantities become random variables

Examples of unknown quantities include:

- a future target y_*;
- a latent representation z;
- model parameters θ;
- an unobserved state in a dynamical system;
- the causal effect of an intervention;
- the return obtained by an action.

A probability distribution encodes which values are plausible and how strongly they are supported.

4.2 Four broad task families

Task	Typical probabilistic object	Core question
Prediction	$p(y \mid x)$	What output is plausible for this input?
Generation	$p(x)$ or $p(x \mid c)$	What observations could be generated, possibly under a condition c?
Discovery	$p(z, x) = p(z)p(x \mid z)$ and $p(z \mid x)$	What hidden structure may explain the observations?
Action	$p(s', r \mid s, a)$ plus a utility or loss	Which action has the best expected consequence?

4.3 A generic latent-variable model

Let θ be global model parameters, z latent variables, and $\mathcal{D}$ observed data. A common factorization is

$$p(\mathcal{D}, z, \theta) = p(\mathcal{D} \mid z, \theta)\, p(z \mid \theta)\, p(\theta).$$

After observing data, Bayes' rule gives

$$p(z, \theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid z, \theta) p(z \mid \theta) p(\theta)}{p(\mathcal{D})}.$$

For a new input x_*, the posterior predictive distribution is obtained by averaging over parameter uncertainty:

$$p(y_* \mid x_*, \mathcal{D}) = \int p(y_* \mid x_*, \theta)\, p(\theta \mid \mathcal{D})\, d\theta.$$

A point-estimate method replaces $p(\theta \mid \mathcal{D})$ by one value $\hat{\theta}$. This is computationally convenient, but it may be overconfident when data are limited or the model is weakly identified.

4.4 Aleatoric and epistemic uncertainty

- **Aleatoric uncertainty** is intrinsic variability in the data-generating process. Even with the correct model and unlimited data, it may remain.
- **Epistemic uncertainty** reflects limited knowledge about the model, parameters, or hidden mechanism. It can often be reduced by collecting informative data or improving the model.

This distinction matters. Collecting more labels is useful when uncertainty is epistemic, but may not reduce irreducible aleatoric noise.

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5. Review I — Probability

5.1 Probability space and random variables

A probability space is a triple $(\Omega, \mathcal{F}, P)$:

- Ω: sample space of possible outcomes;
- $\mathcal{F}$: collection of measurable events;
- P: probability measure satisfying non-negativity, normalization, and countable additivity.

A random variable $X : \Omega \to \mathcal{X}$ maps outcomes to values. Its distribution may be represented by a probability mass function for discrete variables or a probability density function for continuous variables.

5.2 Joint, marginal, and conditional distributions

For random variables X and Y:

$$p(x, y) = p(y \mid x) p(x) = p(x \mid y) p(y).$$

Marginalization removes variables:

$$p(x) = \sum_y p(x, y) \quad \text{or} \quad p(x) = \int p(x, y)\, dy.$$

Conditional independence is written

$$X \perp Y \mid Z \quad \Longleftrightarrow \quad p(x, y \mid z) = p(x \mid z) p(y \mid z).$$

Conditional independence will become the main language of probabilistic graphical models.

5.3 Expectation and variance

For a continuous random variable,

$$\mathbb{E}[X] = \int x\, p(x)\, dx, \qquad \mathbf{Var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right].$$

Two useful identities are

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b,$$

and

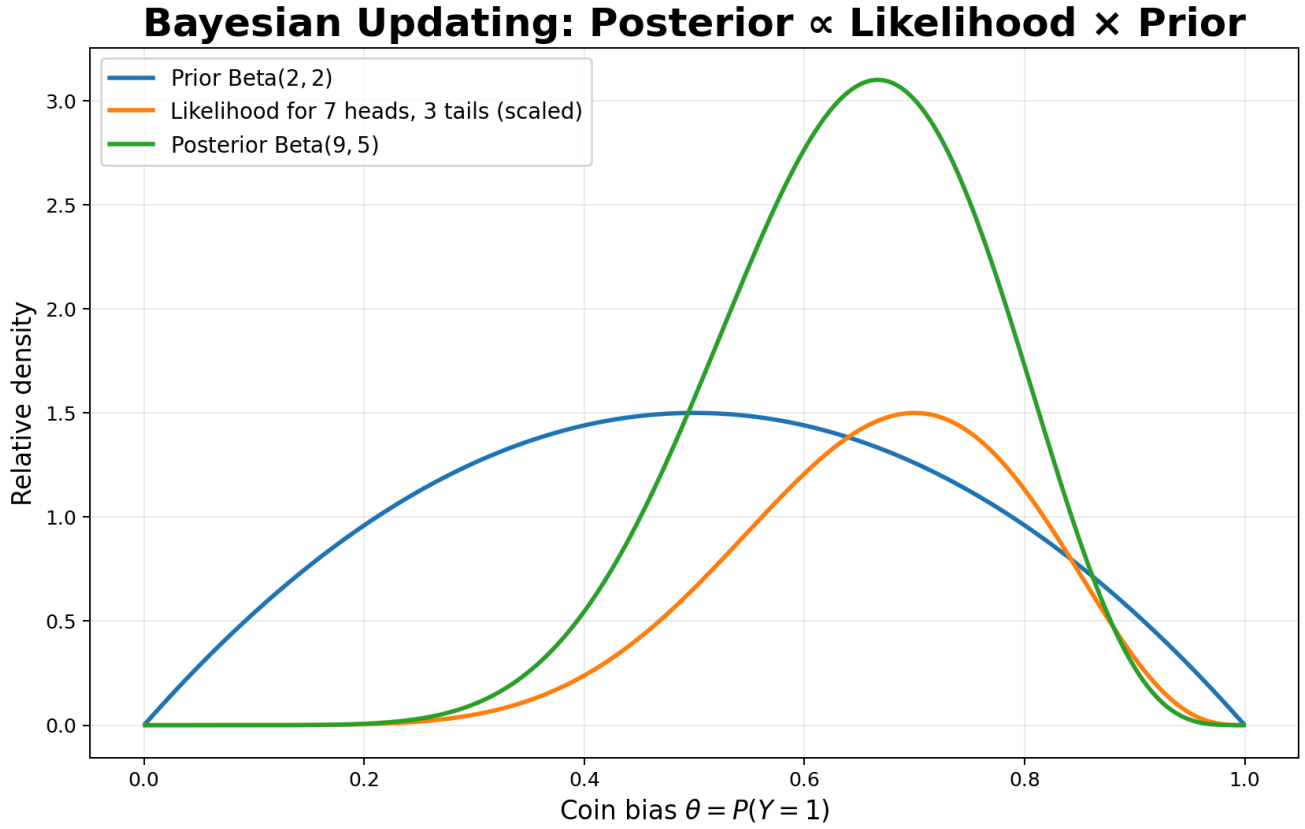
$$\mathbf{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2.$$

Expectation is not merely a descriptive statistic. It defines risks, expected utilities, Monte Carlo estimators, and many learning objectives.

5.4 Bayes' rule

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta) p(\theta)}{p(\mathcal{D})} \propto p(\mathcal{D} \mid \theta) p(\theta).$$

- $p(\theta)$: prior;
- $p(\mathcal{D} \mid \theta)$: likelihood;
- $p(\theta \mid \mathcal{D})$: posterior;
- $p(\mathcal{D})$: marginal likelihood or evidence.



In the figure, a $\mathbf{Beta}(2,2)$ prior is updated with seven heads and three tails. The posterior is $\mathbf{Beta}(9,5)$. The update changes both the location and the concentration of our belief.

Quick check

Suppose H_1 is a fair coin and H_2 is a coin with $P(\text{head}) = 0.8$. Let the prior probabilities be equal. After observing two heads,

$$p(H_2 \mid HH) = \frac{0.8^2 \cdot 0.5}{0.8^2 \cdot 0.5 + 0.5^2 \cdot 0.5} = \frac{0.64}{0.64 + 0.25} \approx 0.719.$$

The posterior is not determined by the likelihood alone; it depends on both likelihood and prior.

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6. Review II — Statistics

Statistics asks how to learn unknown quantities from finite data and how to quantify the resulting uncertainty.

Let

$$\mathcal{D} = \{(x_n, y_n)\}_{n=1}^{N}.$$

Under the iid assumption,

$$p(\mathcal{D} \mid \theta) = \prod_{n=1}^{N} p(y_n \mid x_n, \theta).$$

Taking logarithms converts the product into a sum:

$$\log p(\mathcal{D} \mid \theta) = \sum_{n=1}^{N} \log p(y_n \mid x_n, \theta).$$

6.1 Maximum likelihood estimation

$$\hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} p(\mathcal{D} \mid \theta) = \arg\min_{\theta} \left[-\sum_{n=1}^{N} \log p(y_n \mid x_n, \theta) \right].$$

The minimized expression is the **negative log-likelihood (NLL)**. Many familiar losses are NLLs:

- squared error corresponds to a Gaussian observation model with fixed variance;
- binary cross-entropy corresponds to a Bernoulli model;
- multiclass cross-entropy corresponds to a categorical model.

6.2 Maximum a posteriori estimation and regularization

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} \left[\log p(\mathcal{D} \mid \theta) + \log p(\theta) \right].$$

Equivalently,

$$\hat{\theta}_{\mathrm{MAP}} = \arg\min_{\theta} \left[-\log p(\mathcal{D} \mid \theta) - \log p(\theta) \right].$$

Therefore, a prior can be interpreted as a regularizer:

- Gaussian prior $p(\theta) \propto \exp(-\lambda \|\theta\|_2^2 / 2)$ gives an ℓ_2 penalty;
- Laplace prior $p(\theta) \propto \exp(-\lambda \|\theta\|_1)$ gives an ℓ_1 penalty.

6.3 Point estimation versus Bayesian inference

Method	Representation of uncertainty	Typical computation
MLE	One parameter value	Optimize likelihood
MAP	One parameter value influenced by a prior	Optimize posterior density
Full Bayesian inference	Distribution $p(\theta \mid \mathcal{D})$	Integrate or approximate the posterior

Full Bayesian prediction averages over plausible parameter values. This is often more reliable in small-data regimes, but the required integral is usually intractable. Much of this course is therefore about **approximate inference**.

6.4 Generalization reminder

A model should not only minimize training loss. We care about its expected loss on future data:

$$R(\theta) = \mathbb{E}_{(x,y)\sim p^*} \left[\ell(y, f(x; \theta)) \right].$$

Since the true data distribution p^* is unknown, we estimate performance using validation and test data. Regularization, model selection, calibration, and distribution-shift analysis are all responses to the gap between training performance and future performance.

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7. Review III — Optimization

Learning usually reduces to an optimization problem:

$$\theta^* \in \arg\min_{\theta \in \Theta} L(\theta).$$

7.1 Local and global optima

A global optimum has no worse objective value than any feasible point. A local optimum is only best within a neighborhood. Convexity is important because, for a convex objective, every local minimum is global.

A differentiable function f is convex when

$$f(\lambda x + (1-\lambda)y) \le \lambda f(x) + (1-\lambda) f(y), \qquad 0 \le \lambda \le 1.$$

Deep neural-network objectives are generally nonconvex, so optimization may encounter saddles, flat directions, ill-conditioning, and multiple equivalent solutions.

7.2 Gradient and Hessian

The gradient

$$\nabla L(\theta)$$

points in the direction of steepest local increase. Therefore, $-\nabla L(\theta)$ is a descent direction. The Hessian

$$H(\theta) = \nabla^2 L(\theta)$$

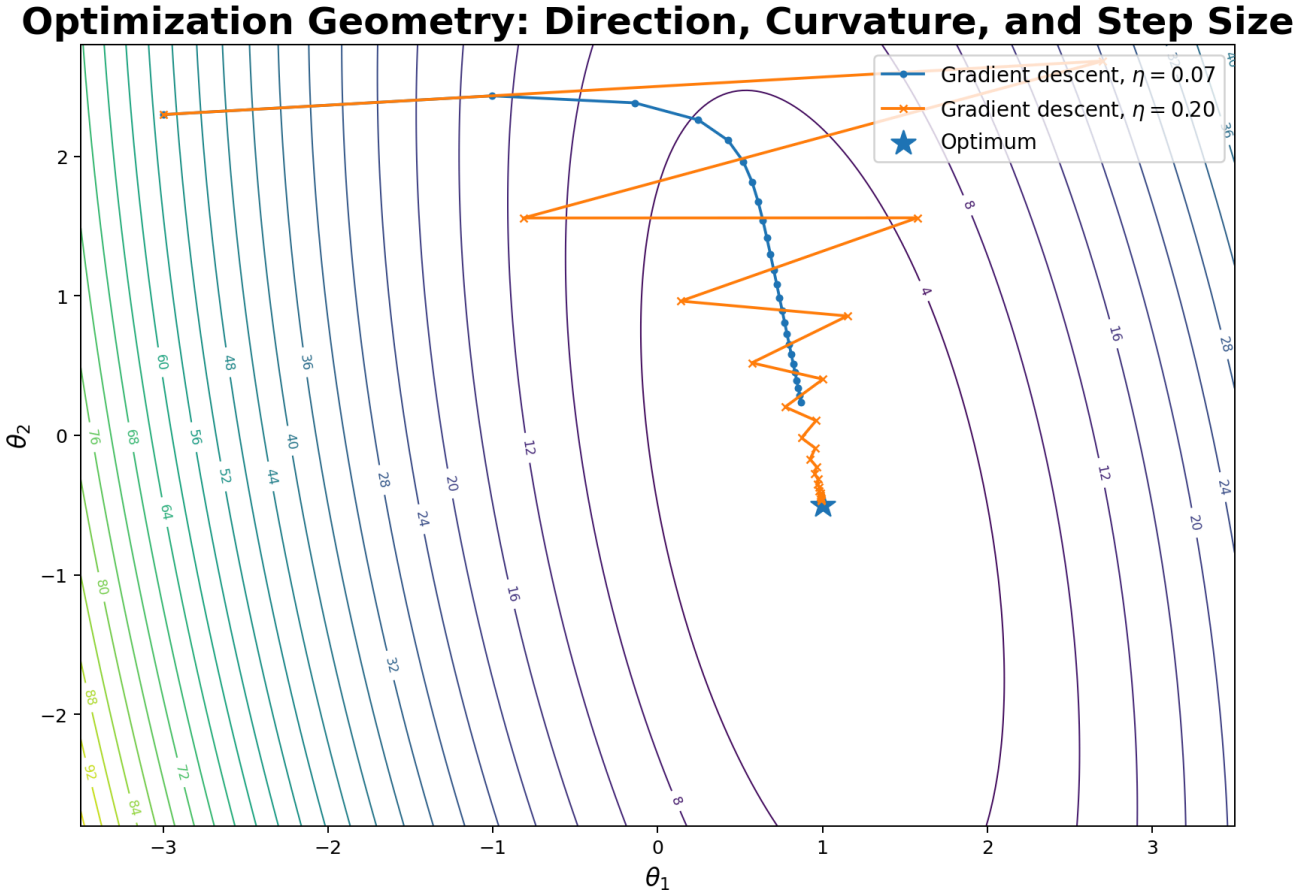
describes local curvature. At a twice-differentiable local minimum, a necessary condition is $\nabla L(\theta) = 0$, and the Hessian is positive semidefinite.

7.3 Gradient descent

$$\theta_{t+1} = \theta_t - \eta_t \nabla L(\theta_t),$$

where $\eta_t > 0$ is the step size or learning rate.

- If the learning rate is too small, convergence is slow.
- If it is too large, the iterates may oscillate or diverge.
- Curvature and conditioning determine how difficult the landscape is to traverse.



7.4 Stochastic gradient descent

For an empirical objective

$$L(\theta) = \frac{1}{N} \sum_{n=1}^{N} \ell_n(\theta),$$

SGD uses a minibatch $\mathcal{B}_t$:

$$\widehat{\nabla L}(\theta_t) = \frac{1}{|\mathcal{B}_t|} \sum_{n \in \mathcal{B}_t} \nabla \ell_n(\theta_t),$$

and updates

$$\theta_{t+1} = \theta_t - \eta_t \widehat{\nabla L}(\theta_t).$$

The minibatch gradient is noisy but inexpensive. This tradeoff makes SGD practical for large datasets and deep models. Later, we will also meet optimization *inside inference*, such as ELBO maximization in variational inference, and sampling methods whose dynamics resemble noisy optimization.

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8. One unifying example: Bayesian logistic regression

For binary classification,

$$p(y_n = 1 \mid x_n, w) = \sigma(w^\top x_n), \qquad \sigma(a) = \frac{1}{1 + e^{-a}}.$$

The Bernoulli negative log-likelihood is binary cross-entropy:

$$-\log p(\mathcal{D} \mid w) = -\sum_{n=1}^{N} \left[y_n \log \sigma(w^\top x_n) + (1 - y_n) \log(1 - \sigma(w^\top x_n)) \right].$$

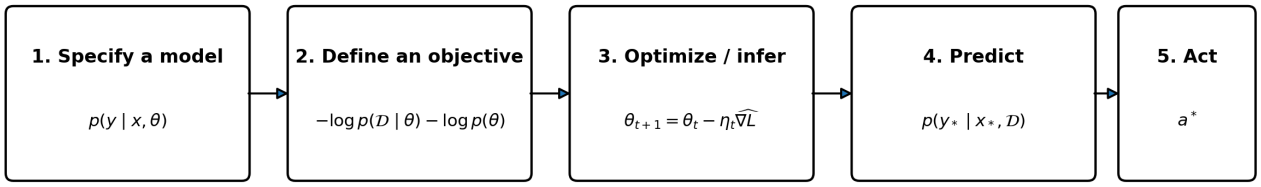
With a Gaussian prior $w \sim \mathcal{N}(0, \lambda^{-1} I)$, MAP estimation minimizes

$$-\log p(\mathcal{D} \mid w) + \frac{\lambda}{2} \|w\|_2^2 + \text{constant}.$$

This single model connects today's three reviews:

1. **Probability:** Bernoulli likelihood and Gaussian prior.
2. **Statistics:** MLE, MAP, posterior, and predictive uncertainty.
3. **Optimization:** gradient-based minimization of the NLL or MAP objective.

Probability → Statistics → Optimization → Prediction → Decision



This chain will reappear in graphical models, variational inference, Bayesian deep learning, generative models, and reinforcement learning.

In later weeks, the same structure becomes more difficult because the posterior is high-dimensional, multimodal, or analytically unavailable. Variational inference, Monte Carlo methods, MCMC, and sequential Monte Carlo are different strategies for coping with that difficulty.

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9. Diagnostic self-check

You should be able to answer the following without extensive calculation.

1. What is the difference between $p(y \mid x)$ and $p(x \mid y)$?

2. Why does independence imply zero covariance, while zero covariance does not generally imply independence?
3. Why is minimizing cross-entropy often equivalent to maximum likelihood estimation?
4. How does a prior become a regularization term in MAP estimation?
5. What is the difference between aleatoric and epistemic uncertainty?
6. Why can a large learning rate cause divergence even for a convex objective?
7. What is lost when $p(\theta \mid \mathcal{D})$ is replaced by a point estimate $\hat{\theta}$?

Compact answers

1. They condition in opposite directions and are related by Bayes' rule, not by symmetry.
2. Covariance captures only linear dependence; nonlinear dependence may remain.
3. Cross-entropy is the NLL for Bernoulli or categorical observation models.
4. The negative log-prior is added to the NLL.
5. Aleatoric uncertainty is intrinsic data variability; epistemic uncertainty reflects limited knowledge.
6. The update can overshoot along directions of high curvature and oscillate or diverge.
7. Parameter uncertainty and model averaging are discarded, often producing overconfident predictions.

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10. Take-home messages

1. **Advanced machine learning is not a list of disconnected algorithms.** It is a set of modeling and inference tools connected by probability, statistics, optimization, and decision theory.
2. **A probabilistic model expresses assumptions about a data-generating process.** Those assumptions determine what can be inferred and how uncertainty is propagated.
3. **MLE, MAP, and Bayesian inference are related but not identical.** MLE and MAP produce point estimates; Bayesian inference retains a distribution over plausible parameters or latent variables.
4. **Optimization is the computational engine of learning.** The geometry of the objective and the quality of gradient estimates strongly affect what solution is reached.
5. **Throughout the semester, always ask four questions:** What is the model? What is observed and hidden? What inference or optimization problem must be solved? How will uncertainty affect prediction or action?

Preparation for the next class

Read or skim:

- Main textbook, Chapter 4, especially §§4.1–4.2 on directed and undirected graphical models;
- Main textbook, Chapter 7, §§7.1–7.4 for the overview of inference algorithms;
- Main textbook, Chapter 9, §§9.1–9.3 for message passing on chains and trees.

Bring a laptop. We will begin using conditional independence and factorization to turn complex joint distributions into computable models.

References and attribution

- Murphy, K. P. (2023). *Probabilistic Machine Learning: Advanced Topics*. MIT Press.
- Murphy, K. P. (2022). *Probabilistic Machine Learning: An Introduction*. MIT Press.
- Murphy, K. P. (2025). *Supplementary Material for Probabilistic Machine Learning: Advanced Topics*.
- Course policy and weekly sequence follow the official Fall 2026 syllabus for DSC1002.
- All diagrams on this page were created specifically for this course. No textbook figure has been modified or reproduced in this package.

Reading the optional code and figures

The figures explain different operations. In the Bayesian update plot, the horizontal axis is an unknown parameter and the posterior combines prior information with observed data. In an optimization contour plot, the axes are parameter coordinates and successive points are algorithm iterates; a visually smooth path is not a posterior distribution. State what is fixed before interpreting either image.

The optional beta-binomial notebook defines `make_graph(data, save_name)`. `data` is a dictionary with prior `a,b`, likelihood `n_0,n_1`, and posterior `a,b`; `save_name` is an output filename. It plots curves and saves a figure, returning no numerical posterior. The caller must supply the consistent posterior parameters `a+n_1,b+n_0`; changing counts alone does not update the supplied posterior. JAX, Matplotlib and probml_utils are required. Likelihood uses a second vertical axis and need not integrate to one over the parameter.

The optional steepest-descent notebook defines `gradient_descent(x0, f, f_prime, hessian, stepsize=None)`. `x0` is a two-coordinate starting point, `f` a scalar objective, `f_prime` its two-entry gradient, and `stepsize` either a fixed scalar or None for SciPy line search. It returns three lists of visited x coordinates, y coordinates and objective values. The passed `hessian` is unused by this routine. The implementation sets the step to zero if line search fails, so an unchanged point alone does not prove convergence. Its plot title says exact line search, but the code calls a Wolfe-condition line search. Create the `figures` directory before its saving cells. These are upstream optional examples, not a claim that every upstream dependency or version has been executed in this course environment.

These usage notes come from the locally preserved upstream notebooks inspected on 2026-09-09. Follow the current upstream setup when running them, and record the actual revision. The course's W2 finite-model package provides a smaller, separately validated inference exercise.